HIGGINS DECOMPOSITION (H^S)

Operationalizing compositional data analysis — a runnable standard for researchers and the AI assistants they choose

“Compositional monitoring of energy-mix drift on the simplex” • CoDaWork 2026 • Coimbra, Portugal • June 1–5

Peter Higgins • Rogue Wave Audio / Binaural Test Lab • Markham, Ontario, Canada

EN · FR · ES · RU · ZH · AR
handout in UN-6 locales

11	101	44	3	22+66	~220
validated domains	reference datasets	orders of magnitude	IEEE-floor physics confirmations	slides — talk + cinema scroll	glossary v3.0 entries

What this is. Aitchison gave the field its geometry in 1986; CoDaWork has trained four decades of methodologists. H^S packages current and developing CoDa methods into a **runnable operational standard** — seven phases, two human gates, deterministic hash-chained output, identical results whether the keystrokes come from a researcher or from an AI assistant. The math is standard; the operational frame may be new.

Why operationalize compositional analysis

Measurability. Turns theoretical compositional structure into reproducible per-step diagnostics: helmsman (handedness), Power Share, Activation Coefficient, navigation bearing. Quantifiable, comparable, audit-able.

Consistency. Fixed schema (CNT v3.1.0 / CNQ v2.0.0), fixed pipeline, byte-identical re-runs across machines, OS, BLAS, and years. Same input, same output, always.

Hypothesis testing. The same engine that produces a finding produces the falsifiability frame (MC-4 — four named defeat paths) that would overturn it. No publication-without-falsification.

Numerical and operational advantages over ad-hoc CoDa pipelines

- **Helmert-ILR orthonormal coordinates** — no choice-of-balance arbitrariness; deterministic basis across teams.
- **atan2 for helmsman and navigation angles** — wrap-safe across ±π; no precision loss or sign jumps at the cycle boundary.
- **Hash-chained provenance** — SHA-256 from raw CSV through CNT JSON, plates, projector, and manuscript figure. A reviewer in 2030 can prove nothing changed.
- **IEEE-floor cross-platform determinism** — same input, same output, every machine, every time. Verified on Backblaze, Planck CMB, neutrino oscillations.
- **Coherent Range Doctrine (CRD-1.0)** — multi-carrier comparisons on intersection of all members' ranges; asymmetric-drift artifacts eliminated.

The three-layer operational standard

CNT v3.1.0 measure	Closure → CLR → Helmert-ILR → per-step compositional metrics, helmsman, Power Share, Activation Coefficient, navigation, diagnostics, hashes. Current source v3.2.0 adds `navigation_2d` for ILR-Helmert PCA barycenter trajectory.
CNQ v2.0.0 name the algebra	Quaternion-view dashboards and higher-order structure diagnostics (CHSH joint coherence, twin-quaternion factoring at D=8 with Tsirelson-bound respect). Algebraic companion to CNT.
CCTT v1.0 operationalize	**The runnable standard.** Seven phases (diagnose → adapter *gate* → engine → outputs → render → self-verify *gate* → present + journal). Two human gates; everything else deterministic. **The repo trains both researcher and AI assistant — same protocol, identical hash-verifiable output.**

Five viewpoints in the talk

- **Composition** — share each carrier has.
- **Helmsman** — largest CLR displacement at a step.
- **Helmsman trajectory** — when steering changes.
- **Power Share** — how much squared CLR motion each carrier did.
- **Activation Coefficient** — Power Share ÷ starting share = "yeast factor."

CCTT 7-phase protocol

1 Diagnose | **2** Adapter *(gate)* | **3** Engine | **4** Outputs | **5** Render | **6** Self-verify *(gate)* | **7** Present + journal

Operational evidence — what the standard surfaces: a carrier can be small in share but large in structural work. **USA Solar 2012 → 2013:** 0.107% starting share, 81.7% of structural Power Share, **Activation Coefficient ≈ 760×.** The cross-country deceptive-drift signature fires in **5 of 9 countries** (AUS, CHN, GBR, IND, JPN) and does *not* fire in DEU (annual), FRA, USA, or WLD. The protocol discriminates; it does not over-fire. **A regression on raw shares would not surface either finding.**

Standard onboarding — choose your entry point

1. **Conference attendee:** `CODA-Association/CONFERENCE_ATTENDEES.md` — slide-by-slide follow-along.
2. **Explore visually (zero install):** `CODA-Association/CODAwork2026/data_outputs/codawork2026_projector.html` .
3. **Run your own composition:** `QUICKSTART.md` + `ai-refresh/CCTT_QUICKSTART.md` — 7-phase runbook, manual or AI-assisted.
4. **Verify a published number:** manuscript + Supplementary Information + per-country JSON + hash chain.
5. **Vocabulary lookup:** `HCI-CNT/handbook/GLOSSARY.md` v3.0 (~220 entries: PCA, SVD, CLR/ILR, Helmert, CHSH, Tsirelson, Activation Coefficient, MC-1..MC-4).

Talk	“Compositional monitoring of energy-mix drift on the simplex,” CoDaWork 2026, Coimbra, June 1–5. Find Peter during sessions and Q&A — happy to walk the projector live.
Contact	Peter Higgins — **PeterHiggins@RogueWaveAudio.com** • Rogue Wave Audio / Binaural Test Lab, Markham, Ontario, Canada
Repository	github.com/PeterHiggins19/higgins-decomposition (QR top-right) • community folder: `CODA-Association/` • conference folder: `CODA-Association/CODAwork2026/`
How to cite	Higgins, P. (2026). Compositional monitoring of energy-mix drift on the simplex. CoDaWork 2026, Coimbra, Portugal. Repo: github.com/PeterHiggins19/higgins-decomposition (commit in `HS_FAST_REFRESH.json`).
How to adopt	Fork the repo, run the 7-phase CCTT on your composition, file a `JOURNAL.md` . AI assistant follows the same gates. See `ai-refresh/COMMUNITY_TEST_PACKET.json` for the structured adoption test.
License	Apache-2.0 (code) • CC BY 4.0 (docs and figures). Fully open source — fork, audit, extend, attribute.

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line. The AI follows the same protocol.

Side 2 — Operations, symbols, and the apparatus map

Compact reference for the operations performed by CoDa methods, what H^s adds on top, what CNQ adds in the quaternion view, where the closure comes from in each domain, and which apparatus reads what.

CoDa core operations (the foundation Aitchison gave the field in 1986)

Operation	Symbol	Formula / definition
Closure	$C(x)$	$x / \Sigma_i x_i$
Geometric mean	$g(x)$	$(\prod_i x_i)^{(1/D)}$
CLR (centred log-ratio)	$clr_i(x)$	$\log(x_i) - (1/D)\ \Sigma_j \log(x_j)$
ILR (Helmert)	$\eta(x)$	$V^T \cdot clr(x),\ V \cdot V^T = I$
Aitchison distance	$d_{Ait}(x,y)$	$\|clr(x) - clr(y)\|_2$
Perturbation	$x \oplus y$	$C(x \odot y)$ — additive on the simplex
Power scaling	$\alpha \odot x$	$C(x^\alpha)$ — scalar action on the simplex

H^s supplementary operations (what the standard adds)

Operation	Symbol	Formula / definition
Helmsman index	$\sigma(t)$	$\mathrm{argmax}_i\ \lvert clr_i(t+1) - clr_i(t)\rvert$
Aitchison-step	$\|\Delta clr(t)\|$	$\|clr(t+1) - clr(t)\|_2$
Power Share	$\pi_j(t)$	$(\Delta clr_j)^2 / \Sigma_k (\Delta clr_k)^2,\ \Sigma \pi_j = 1$
Activation Coefficient	$\alpha_j(t)$	$\pi_j(t) / \rho_j(t)$ (when $\rho_j \geq 10^{-3}$)
Shannon entropy	$H(t)$	$-\Sigma_j \rho_j \ln \rho_j$
Effective carriers	$K_{eff}(t)$	$\exp(H(t))$
L2 drift	$L2(p,q)$	$\sqrt{\Sigma_i (p_i - q_i)^2}$
TV distance	$TV(p,q)$	$(1/2)\ \Sigma_i \lvert p_i - q_i\rvert$

CNQ quaternion operations (the phase readout on S^3)

Operation	Symbol	Formula / definition
Phase quaternion	$q(t)$	$\in S^3 \cong SU(2)$
Quaternion conjugate	q^*	$(a, -b, -c, -d)$
Hamilton product	$(p \cdot q)_k$	non-commutative quaternion multiplication
Quaternion sandwich	v'	$q \cdot v \cdot q^*$ (rotation of 3-vector)
Quaternion log	$\log(q)$	$(\mathrm{atan2}(\lvert v\rvert, a) / \lvert v\rvert) \cdot v$
Metric involution	M^2	$= I \Leftrightarrow (q^*)^* = q$
SLERP (spherical interp)	$slerp(q_1,q_2,\alpha)$	$\sin((1-\alpha)\Omega)/\sin\Omega \cdot q_1 + \sin(\alpha\Omega)/\sin\Omega \cdot q_2$
CHSH joint coherence	S	$E(a,b) + E(a,b') + E(a',b) - E(a',b')$

Closure constraints across domains (the budget the partition apportions)

Domain	Budget	Closure constraint
Acoustic (BTL)	$c = 20 \cdot \log_{10}(2) \approx 6.02$ dB	$\Sigma G_i = c$ ($4\pi \to 2\pi$ baffle-step)
Electrical mix	100 % generation	$\Sigma p_i = 1$ (coal+gas+hydro+nuclear+solar+wind+oil+other)
Geochemistry	100 % weight	$\Sigma w_i = 1$ (major-element oxide fraction)
Macro-economic	100 % GDP	$\Sigma p_i = 1$ (sectoral share)
ERB loudness (HCI-AUDIO)	100 % perceptual	$\Sigma_j \Sigma_{drivers} = 1$ (40 bands × 4 drivers)

Apparatus at a glance — who reads what

Apparatus	Reads	Output
CoDa community	static partition / one timestep	log-ratio biplot, distance matrix
CNT — tensor engine	trajectory amplitude / per-timestep	simplex coords, Helmsman, Power Share, navigation
CNQ — quaternion engine	phase trajectory / S^3 rotation rates	quaternion path, CHSH, twin-quaternion factoring
HCI-AUDIO	4-way listening-position field	ERB band × driver matrix, phase quaternions
HUF (umbrella)	governance	HUF-STD-001 (Publication), -002 (Tensor Train I/O), -003 (Linear Algebra Foundations)

HUF-STD-002 Tensor Train — pipeline order, link, mode, rank

Order	Link	Mode (input → output)	Rank
Order 0	Adapter	raw → CSV (T × D)	D = 2 … 9+
Order 1	CNT — closure + Helmert-ILR	(T, D) → (T, D − 1)	D − 1
Order 2	CNT — per-step viewpoints	(T, D − 1) → (T, K)	K = 5 metrics
Order 3	CNT — depth tower + IR class	(T, K) → scalar block	regime label
Order 2-3	CNQ — quaternion path	CNT JSON → (T, 4) at D = 2 / 3 / 4	4 ($S^3 \cong SU(2)$)
Order 4	Vector render	JSON → plate tensor	PDF · PNG · SVG

Flow: raw → [Adapter] → CSV → [CNT v3.1.0] → cnt_*.json → [CNQ v2.0.0] → cnq_*.json → [Render] → PDF · PNG · SVG

K = 5 metrics : Helmsman · Aitchison-step · Power Share · Activation Coefficient · navigation_2D · Each link emits SHA-256; chain reproducible from raw input to final artifact in one command.

Symbols legend

D carriers · **T** timesteps · p_i portion · G_i gain (dB) · F_c cutoff · τ group delay · $\hat{n}$ rotation axis · **q** unit quaternion · σ Helmsman · α_j Activation · π_j Power Share · η ILR coordinate · **clr** centred log-ratio · $g(x)$ geometric mean · $S^{(D-1)}$ simplex · S^3 3-sphere ≅ SU(2)

Same input, same output, always. The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line. The AI follows the same protocol.